

04. Question Bank & Scenario Repository – Sr. Automation Engineer / Sr. SDET



Page is about?

This section provides a **centralized repository** of level-specific interview questions across all major skill areas. It ensures consistent and thorough technical evaluation aligned with the job expectations for Sr. SDET / Sr. Automation Engineer roles.

Purpose

- Enable consistent technical screening across all panelists
- Categorize questions by **skill area** and **difficulty level**
- Include practical exercises, real-world scenarios, and judgment-based situations
- Promote reusable, role-aligned assessments

Question Categories

1. Java (Basic Advanced)

Basic Java Questions

Question	Expected Answer / Notes
What is the difference between == and .equals() in Java?	== checks reference equality, .equals() checks value equality
What is a constructor? How is it different from a method?	Initializes an object, no return type
What are the access modifiers in Java?	private, default, protected, public
What is method overloading vs overriding?	Overloading: same method name, diff params. Overriding: runtime polymorphism
Explain final, finally, and finalize()	final (constant), finally (block), finalize() (GC hook)

Intermediate Java Questions

Question	Expected Answer / Notes
Explain HashMap internal working.	Uses hashCode & equals for bucketing
Difference between ArrayList and LinkedList?	Access time, insertion/removal performance
How is exception handling managed in test code?	try-catch, custom exceptions, assertions
Interface vs Abstract class?	Interface = behavior, Abstract = partial implementation
What is immutability in Java?	Object state cannot be changed once created
What is the difference between a class and an object in Java?	Class is a blueprint; object is an instance of the class.
What is the default value of boolean in Java?	FALSE
Can a constructor be private in Java?	Yes, often used in Singleton pattern.
What are static variables and methods?	Shared across all instances of a class.
What is the this keyword?	Refers to the current instance of the class.
What is a package in Java?	A namespace that organizes classes and interfaces.
What is a wrapper class?	Converts primitives to objects (e.g., int Integer).
Difference between StringBuffer and StringBuilder?	StringBuilder is faster but not thread-safe.
How is Array different from ArrayList?	Arrays are fixed-size; ArrayLists are resizable.

What is autoboxing?	Automatic conversion between primitives and wrapper classes.
What is a ternary operator?	Shortcut for if-else: condition ? expr1 : expr2.
How to prevent a class from being subclassed?	Declare it as final.
What is method hiding?	Static methods with the same signature in child classes.
What is JVM, JRE, JDK?	JVM: runs Java, JRE: JVM + libs, JDK: JRE + tools.
What is a do-while loop?	Executes loop body first, then checks the condition.
What does super keyword do?	Refers to the immediate parent class.
What are logical operators?	&&, ^
What is an infinite loop?	A loop with a condition that never becomes false.
What is the difference between break and continue?	break exits the loop; continue skips iteration.
What are primitive data types in Java?	byte, short, int, long, float, double, char, boolean

Advanced Java Questions

Question	Expected Answer / Notes
What are functional interfaces and lambda expressions?	One-method interfaces; Lambda = concise function representation
What is stream API and how do you use <code>filter()</code> or <code>map()</code> ?	Process collections declaratively
How does garbage collection work in Java?	Mark-and-sweep, GC tuning concepts
Multithreading: Explain synchronized and deadlock prevention	Locking mechanism and best practices
Use-case based: Write code to group a list of strings by their first character using streams.	Expected: <code>Collectors.groupingBy()</code>

2. Selenium + Framework (TestNG / Cucumber)

Selenium WebDriver

Question	Expected Answer
How do you handle dynamic web elements?	Use relative XPath, custom waits
Explain Page Object Model.	Separation of locators and logic
How do you handle popups and alerts?	<code>driver.switchTo().alert()</code>
How do you take screenshots on test failure?	<code>ITestListener</code> or Hooks
Implicit vs Explicit Waits?	Implicit = global, Explicit = conditional wait
What are the different types of waits in Selenium?	Implicit, Explicit, Fluent Wait
How do you handle iframe in Selenium?	<code>driver.switchTo().frame()</code>
What is the use of Actions class?	For advanced mouse and keyboard actions
How to verify tooltip text in Selenium?	Get attribute title or hover using Actions
What are the different types of locators in Selenium?	ID, Name, ClassName, XPath, CSSSelector, etc.
How do you handle dropdowns in Selenium?	Using Select class (<code>selectByVisibleText()</code>)
How to manage test data in Selenium tests?	Using Excel/JSON or TestNG <code>@DataProvider</code>
How to check for broken links?	Use <code>URLConnection</code> in Java
How do you scroll a page in Selenium?	Use <code>JavascriptExecutor</code> to scroll
What is DesiredCapabilities?	Set browser properties before execution
What is headless testing?	Test execution without opening a browser UI

How to check if element is visible/enabled?	isDisplayed(), isEnabled(), isSelected()
What is Page Factory?	Enhances POM using @FindBy annotations
Difference between driver.close() and driver.quit()?	close() closes current tab; quit() closes all tabs
How do you handle alerts with input boxes?	Use alert.sendKeys()
What are TestNG annotations?	@BeforeMethod, @Test, @AfterMethod, etc.
How do you group test cases in TestNG?	Use groups attribute in @Test
How to rerun failed test cases in TestNG?	Use IRetryAnalyzer or rerun suite
What is soft assertion in TestNG?	Doesn't stop execution after failure
What is parallel execution in TestNG?	Run tests concurrently using threadPoolSize

TestNG / BDD Framework

Question	Expected Answer
Difference between TestNG & JUnit?	TestNG offers more annotations, parallel execution
How do you use DataProvider in TestNG?	Pass dynamic data to test methods
How is glue code structured in Cucumber?	Step defs hooks runners
Explain tagging strategy.	Use @Smoke, @Regression, etc.
How do you generate BDD reports?	Use plugins like Allure, ExtentReports

3. API Testing – Postman + RestAssured

Postman

Question	Expected Answer
How do you chain requests in Postman?	Save response values in environment variables
How do you validate API response code or body?	Use tests[] section with assertions
How do you pass dynamic values to a request?	Use {{variable}} with environment
How do you debug failed API calls?	Check Headers, Logs, Raw response

RestAssured

Question	Expected Answer
How do you write a GET/POST request in RestAssured?	Use given().header().when().get().then() chain
How do you pass JSON payload from external file?	FileReader or Map<String, Object>
How do you validate response schema?	matchesJsonSchemaInClasspath()
How do you handle token-based authentication?	Save token from one call, pass in header of next

4. SQL + Oracle DB

Question	Expected Answer
Write a query to fetch all employees with salary > average salary.	Subquery with AVG(salary)
Difference between INNER JOIN and LEFT JOIN?	INNER = common rows, LEFT = all left rows + match
How do you compare data across two tables?	Use JOIN + WHERE or MINUS
Query to find duplicate emails in user table?	GROUP BY email HAVING COUNT(*) > 1
Top 3 salaries per department?	Use DENSE_RANK() with PARTITION BY dept

1. What is the difference between INNER JOIN and LEFT JOIN?	INNER JOIN returns only matching records; LEFT JOIN returns all from the left table + matching from the right.
2. Write a query to fetch employees with salary greater than average.	<code>SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);</code>
3. How to find duplicate records in a column?	<code>SELECT column, COUNT(*) FROM table GROUP BY column HAVING COUNT(*) > 1;</code>
4. Write a query to find 2nd highest salary.	<code>SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);</code>
5. What is a subquery vs correlated subquery?	Subquery runs once; correlated subquery runs for each row of the outer query.
6. What is the purpose of ROWNUM?	Used to limit the number of rows returned (e.g., <code>ROWNUM <= 5</code>).
7. How do you retrieve only unique values from a column?	Use <code>SELECT DISTINCT column FROM table;</code>
8. How to get employees who don't belong to any department?	<code>SELECT * FROM employees WHERE dept_id IS NULL;</code>
9. How to get current system date in Oracle?	<code>SELECT SYSDATE FROM dual;</code>
10. What's the difference between DELETE, TRUNCATE, and DROP?	DELETE = row-wise, TRUNCATE = all rows (faster), DROP = removes table structure.
11. Write a query to update a record based on a condition.	<code>UPDATE employees SET salary = salary + 1000 WHERE emp_id = 101;</code>
12. How to join three tables using SQL?	Use chained joins: <code>A JOIN B ON cond JOIN C ON cond</code>
13. What is NVL () in Oracle SQL?	Replaces NULL with a default value: <code>NVL(col, 'default')</code>
14. Difference between HAVING and WHERE?	WHERE filters rows before aggregation; HAVING filters groups.
15. How to calculate total salary grouped by department?	<code>SELECT dept_id, SUM(salary) FROM employees GROUP BY dept_id;</code>
16. How to fetch top N salaries using ROWNUM?	<code>SELECT * FROM (SELECT * FROM employees ORDER BY salary DESC) WHERE ROWNUM <= N;</code>
17. What is a view? How is it different from a table?	A virtual table based on a query; doesn't store data itself.
18. How to get employee details who joined in the last 30 days?	<code>WHERE join_date >= SYSDATE - 30</code>
19. How to check if a table has any rows?	<code>SELECT COUNT(*) FROM table;</code> or <code>EXISTS (SELECT 1 FROM table)</code>
20. What are constraints in SQL? Name a few.	Rules on data: NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK

5. Agile / JIRA / Git / CI-CD

Question	Expected Insight
How do you write a JIRA story for automation?	Clear title, ACs, link to test coverage
What's your defect life cycle from QA perspective?	Log Triage Dev fix Retest Close
Explain Git branch strategy you follow.	Feature Dev QA Release
How do you trigger Jenkins job post-commit?	Git hook or Jenkins webhook
What's your tagging strategy for test execution?	@Smoke, @Regression, @FeatureName

6. Problem-Solving + Scenarios

Scenario	What to Assess
Developer disagrees with a bug marked by QA	Conflict handling, justification
You find regression after a patch is applied	Risk mitigation, RCA
Sprint ends but QA is incomplete	Communication, flexibility, planning
Test is flaky in pipeline	Debugging, isolation of failure

API fails intermittently	Retry logic, backend understanding
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7. Coding + Test Case Exercises

Java Code Tasks

- Write code to reverse words in a sentence without using library
- Find first non-repeating char in string
- Group anagrams from a string array
- Count frequency of all characters using Map
- Detect palindrome and return true/false

Test Automation Scenario

Scenario	What to Automate
Go to Google > Search "automation testing" > Click first result > Validate page title	Validates wait, XPath, validation
Login Form > Valid username/password > Dashboard load	Validates basic form handling
Table with dynamic rows > Find and click checkbox where name = "John"	Validates locator strategy
Upload file > Validate upload complete message	Demonstrates input[type=file] handling

Practice SQL Exercises (with Mock Table Structures)

These exercises simulate real-world test scenarios where automation or SQL validation may be needed.

Mock Schema

Table: employees

Column Name	Data Type	Description
emp_id	NUMBER	Primary key
emp_name	VARCHAR2(100)	Employee full name
dept_id	NUMBER	Foreign key from departments
salary	NUMBER(8,2)	Monthly salary
join_date	DATE	Date of joining

Table: departments

Column Name	Data Type	Description
dept_id	NUMBER	Primary key
dept_name	VARCHAR2(100)	Department name
location	VARCHAR2(100)	Department location

SQL Practice Exercises

#	Question	Expected Output / Goal
1	List all employees with salary > 50,000.	Identify high earners.
2	Count the number of employees in each department.	Validate team distribution.
3	Fetch employee(s) with the maximum salary in each department.	Use of RANK() or MAX() + PARTITION.
4	Get the total salary by department.	Use GROUP BY dept_id.
5	Find all departments without employees.	Use LEFT JOIN + WHERE emp_id IS NULL.

6	List employees who joined in the last 60 days.	Filter using SYSDATE - 60.
7	List employees in "Engineering" department.	Use JOIN on dept_name.
8	Show department-wise average salary.	GROUP BY dept_id + AVG(salary).
9	Show all employees and their department names.	INNER JOIN example.
10	Show employees whose names start with 'A' and end with 'n'.	Pattern matching with LIKE.